

Answer to the q. no. - 01

Multiplicand = 13 = 1101

No. of iteration = 5

Multiplier = 17 = 10001

Iteration	Multiplicand	Product
0	01101	00000 10001
1	01101	01101 10001 00110 11000
2	01101	00011 01100
3	01101	00001 10110
4	01101	00000 11011
5	01101	01101 11011 00110 11011

Result = 00110 11011

Answer to the q. no. - 02

Given,

$$0 \times \text{ABB9609} = \underbrace{1}_{\substack{\downarrow \\ \text{Sign} \\ \text{bit}}} \underbrace{0101011}_{\substack{\text{Biased} \\ \text{exponent}}} \underbrace{1011100101100001001}_{\text{fraction field}}$$

Bias exponent bits = 7

$$\text{Bias} = 2^{7-1} - 1$$

$$= 2^6 - 1$$

$$= 63$$

$$\begin{aligned} \text{Biased exponent} &= 0101011 \\ &= 43 \end{aligned}$$

$$\begin{aligned} \therefore \text{Actual Exponent} &= 43 - 63 \\ &= -20 \end{aligned}$$

$$\therefore \text{Sign bit} = 1$$

$$\text{Fraction field} = 0. \overset{-1}{1} \overset{-3}{0} \overset{-4}{1} \overset{-5}{1} \overset{-6}{0} \overset{-7}{0} \overset{-8}{1} \overset{-9}{1} \overset{-10}{0} \overset{-11}{1} \overset{-12}{0} \overset{-13}{0} \overset{-14}{0} \overset{-15}{0} \overset{-16}{0} \overset{-17}{0} \overset{-18}{0} \overset{-19}{0} \overset{-20}{1}$$

$$= 0.7241297$$

Now,

$$\text{Decimal value} = (-1)^{\text{sign}} \times (1 + \text{fraction}) \times 2^{\text{exponent}}$$

$$= -1.7241297 \times 2^{-20}$$

$$= -1.644258 \times 10^{-6}$$

(Ans)

Answer to the q. no. - 03

Given,

70.78955

→

$$70 = 1000 \ 110$$

$$.78955 = 1100 \ 1010$$

$$\therefore 70.78955 = 1000 \ 110.1100 \ 1010$$

$$= 1.000 \ 1011 \ 0010 \ 10 \times 2^6$$

↓
A.E

$$.78955 \times 2 = 1.579$$

$$.579 \times 2 = 1.158$$

$$.158 \times 2 = 0.316$$

$$0.316 \times 2 = 0.632$$

$$0.632 \times 2 = 1.264$$

$$.264 \times 2 = 0.528$$

$$0.528 \times 2 = 1.056$$

$$0.056 \times 2 = 0.112$$

Biased exponent bit = 9

Here,

$$\text{Bias} = 2^{9-1} - 1$$

$$= 2^8 - 1$$

$$= 255$$

Sign bit = 0

$$\text{Biased exponent} = 255 + 6$$

$$= 261$$

$$= 1000 \ 0010 \ 1$$

$$\text{fraction field} = 36 - (9+1) = 26$$

Now,

0 | 1000 0010 1 | 0001 1011 0010 1000 0000 0000 0000 0000

sign bit

B.E.

F. field.

$$= 0 \times 41466A.000$$

(Ans)

Answer to the q. no. - 04(a)

Given,

$$50.7869 + 79.83 - 29.58$$

Here,

$$50 = 110010$$

$$.7869 = .11001001$$

→

$$50.7869 = 110010.11001001$$

$$= 1.1001011001001 \times 2^5$$

and,

$$79 = 1001111$$

$$.83 = 11010100$$

→

$$79.83 = 1001111.11010100$$

$$= 1.0011111010100$$

$$\times 2^6$$

$$.7869 \times 2 = 1.574$$

$$.574 \times 2 = 1.148$$

$$.148 \times 2 = 0.296$$

$$.296 \times 2 = 0.592$$

$$0.592 \times 2 = 1.184$$

$$.184 \times 2 = 0.368$$

$$.368 \times 2 = 0.736$$

$$0.736 \times 2 = 1.472$$

$$.83 \times 2 = 1.66$$

$$.66 \times 2 = 1.32$$

$$.32 \times 2 = 0.64$$

$$.64 \times 2 = 1.28$$

$$.28 \times 2 = 0.56$$

$$0.56 \times 2 = 1.12$$

$$0.12 \times 2 = 0.24$$

$$0.24 \times 2 = 0.48$$

and,

$$29 = 11101$$

$$.58 = 100110100$$

$$29.58 = 11101.100110100$$

$$= 1.1101100110100$$

$$\times 2^4$$

$$.58 \times 2 = 1.16$$

$$.16 \times 2 = 0.32$$

$$.32 \times 2 = 0.64$$

$$.64 \times 2 = 1.28$$

$$.28 \times 2 = 0.56$$

$$.56 \times 2 = 1.12$$

$$.12 \times 2 = 0.24$$

$$0.24 \times 2 = 0.48$$

Now,

$$50.7869 + 79.83$$

$$= 1.1001101100100 \times 2^5 + 1.00111110101 \times 2^6$$

$$= 0.1100101100101 \times 2^6 + 1.00111110101 \times 2^6$$

$$= (0.1100101100101 + 1.00111110101) \times 2^6$$

$$= 10.0000101011101 \times 2^6$$

$$= 1.000001010011101 \times 2^7$$

$$\begin{array}{r} 1.00111110101 \\ 0.1100101100101 \\ \hline 10.0000101011101 \end{array}$$

Again,

$$(Sum) \times - 29.58$$

$$= 1.0000 \ 0101 \ 0011 \ 101 \times 2^7 - 1 \ 1101 \ 1001 \ 01 \times 2^7$$

$$= 1.0000 \ 0101 \ 0011 \ 101 \times 2^7 - 0.0011 \ 1011 \ 0010 \ 1 \times 2^7$$

$$= (1.0000 \ 0101 \ 0011 \ 101 - 0.0011 \ 1011 \ 0010) \times 2^7$$

$$= 0.1100 \ 1010 \ 0001 \ 101 \times 2^7$$

$$= 1.1001 \ 0100 \ 0011 \ 01 \times 2^6$$

(Ans)

(b)

Given,

$$64.2486 * 49.1832$$

Here,

$$64 = 1000\ 000$$

$$.2486 = .0011\ 1111$$

$$64.2486 = 1000\ 000.0011\ 1111$$

$$= 1.0000\ 0000\ 1111\ 11$$

$$\times 2^6$$

and,

$$49 = 1100\ 01$$

$$.1832 = 0010\ 1110$$

$$49.1832 = 1100\ 01.0010\ 1110$$

$$= 1.1000\ 1001\ 0111\ 0 \times 2^5$$

Now,

$$= (1.0000\ 0000\ 1111\ 11 \times 2^6$$

$$* 1.1000\ 1001\ 0111\ 0 \times 2^5)$$

$$= (1.0000\ 0000\ 1111\ 11 * 1.1000\ 1001\ 0111\ 0) 2^{6+5}$$

$$= 1.1000\ 1010\ 1111\ 0011 \times 2^{11}$$

(Ans)

$$.2486 \times 2 = 0.4972$$

$$0.4972 \times 2 = 0.9944$$

$$.9944 \times 2 = 1.9888$$

$$.9888 \times 2 = 1.9776$$

$$.9776 \times 2 = 1.9552 \downarrow$$

$$.9552 \times 2 = 1.9104$$

$$.9104 \times 2 = 1.8208$$

$$.8208 \times 2 = 1.6416$$

$$.1832 \times 2 = 0.3664$$

$$.3664 \times 2 = 0.7328$$

$$.7328 \times 2 = 1.4656$$

$$.4656 \times 2 = 0.9312$$

$$.9312 \times 2 = 1.8624$$

$$.8624 \times 2 = 1.7248$$

$$.7248 \times 2 = 1.4496$$

$$.4496 \times 2 = 0.8992$$

Answer to the q. no. - 05

Given,

$$28.4810 - (-4.0210)$$

$$= 28.4810 + 4.0210$$

Here,

$$28 = 11100$$

$$0.4810 = .01111011$$

$$28.4810 = 11100.01111011$$

$$= 1.110001111011 \times 2^4$$

and,

$$4 = 100$$

$$.0210 = 00000101$$

$$4.0210 = 100.00000101$$

$$= 1.0000000101 \times 2^2$$

$$0.4810 \times 2 = 0.962$$

$$.962 \times 2 = 1.924$$

$$.924 \times 2 = 1.848$$

$$.848 \times 2 = 1.696$$

$$.696 \times 2 = 1.392$$

$$.392 \times 2 = 0.784$$

$$.784 \times 2 = 1.568$$

$$0.568 \times 2 = 1.136$$

$$.0210 \times 2 = 0.042$$

$$0.042 \times 2 = 0.084$$

$$0.084 \times 2 = 0.168$$

$$.168 \times 2 = 0.336$$

$$.336 \times 2 = 0.672$$

$$.672 \times 2 = 1.344$$

$$.344 \times 2 = 0.688$$

$$.688 \times 2 = 1.376$$

Now,

$$\begin{aligned}
 & 1.1100 \ 0111 \ 1011 \times 2^4 + 1.0000 \ 0001 \ 0111 \times 2^2 \\
 &= 1.1100 \ 0111 \ 1011 \times 2^4 + 0.0010 \ 0000 \ 0010 \times 2^4 \\
 &= (1.1100 \ 0111 \ 1011 + 0.0010 \ 0000 \ 0010) \times 2^4 \\
 &= 10.1110 \ 0111 \ 1011 \times 2^4 \\
 &= 1.0000 \ 0100 \ 0000 \times 2^5 \quad (\text{Ans})
 \end{aligned}$$

Again,

$$\begin{aligned}
 \text{Bias} &= 2^{8-1} - 1 \\
 &= 2^7 - 1 \\
 &= 127
 \end{aligned}$$

$$\begin{aligned}
 \text{Biased exponent} &= \frac{5}{4} + 127 \\
 &= 132
 \end{aligned}$$

$$\begin{aligned}
 \text{Range} &= 0 \text{ to } 2^8 - 1 \\
 &= 0 \text{ to } 255 \\
 &= 1 \text{ to } 254
 \end{aligned}$$

Here,

$$1 \leq 132 \quad \text{and} \quad 132 \leq 254$$

∴ No overflow or underflow.

Answer to the q. no. - 06

Given,

$$\begin{aligned}
 & 0.000101 \times 2^{-85} * 10.1 \times 2^{-90} \\
 &= 1.01 \times 2^{-85-4} * 1.01 \times 2^{-90+1} \\
 &= 1.01 \times 2^{-89} * 1.01 \times 2^{-89} \\
 &= (1.01 * 1.01) \times 2^{-89-89} \\
 &= 1.1001 \times 2^{-178}
 \end{aligned}$$

Here,

$$\text{Bias exponent bits} = 18 - (12 + 1) = 5 \text{ bit}$$

$$\text{Bias exponent} = 45 - 178 = -163$$

$$\text{Range} = 0 \text{ to } 2^5 - 1$$

$$= 0 \text{ to } 31$$

$$= 1 \text{ to } 30$$

$$\therefore -163 < 1$$

There will be underflow.

Answer to the q. no. - 07

Here, $g \leftarrow x_{10}$, $h \leftarrow x_{11}$, $i \leftarrow x_{12}$, $j \leftarrow x_{13}$

CODE :-

flw f_1 , $0[x_{10}]$

flw f_2 , $0[x_{11}]$

flw f_3 , $0[x_{12}]$

flw f_4 , $0[x_{13}]$

fadds f_5 , f_1 , f_2

fadd.s f_6 , f_3 , f_4

fsub.s f_7 , f_5 , f_6

fsw f_7 , $0[x_{14}]$

Answer to the q. no. - 08(a)

Bias is added to the actual exponent in the floating point representation in order to convert signed number to unsigned number. As a result, the hardware can treat the exponent as a positive integer. It will then simplify the operations.

When bias is added to the ~~bias~~ positive exponent the exponent value becomes greater than the bias.

Again, when bias is added to the negative exponent the sign becomes positive and the value becomes less than the bias.

(b)

In optimized multiplication there is no requirement for of any extra register which is used in traditional long & multiplication. Also, ~~the~~ the multiplier is added to the product's msb which is not for long multiplication. We have to perform only right shift to product and add multiplier when required in optimized multiplication. So, it reduces the number of operation in optimized multiplication. Therefore, ~~the~~ ~~is~~ ~~optimi~~ the efficiency and performance gets improved in optimized multiplication than compared to long multiplication.